

House Plan Package — Mandaue, Cebu

Design C (ground floor) · proposed 2nd floor · mono-pitch roof with parapet & solar · elevations
Prepared for Ariel · compiled September 29, 2026 · all dimensions in centimeters unless noted

Lot ≈ 63 sqm

Ground floor ≈ 36 sqm indoor

2nd floor ≈ 36 sqm

8 × 600 W solar (4.8 kW)

Front faces 199° SSW

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Site & orientation

- Location: Mandaue, Central Visayas (≈10°22'N, 123°56'E, 50 m elevation)
- Front / yard faces **199° (SSW)** · road side faces 289° (WNW) · back faces 19° (NNE)
· CR side faces 109° (ESE)
- Neighbors on the east (CR side) and north (back): no windows on those walls
- 2.2 m road on the west side; 370 cm sliding gate into the front yard

Concept drawings developed from hand sketches. Not for construction — to be verified by a licensed architect / structural engineer and checked against the lot survey and local building permit requirements (PD 1096, NSCP).

1 • Summary & areas

Areas are approximate, from the sketch dimensions.

Ground floor — Design C		
Space	Size (cm)	Area
Room (parents / guest)	397 × 220	8.7 sqm
Living room (incl. stairs) — unchanged	380 × 470	16.6 sqm
Dining (front) + kitchen (back)	180 × 470	8.5 sqm
CR (extends 60 cm past living wall)	120 × 210	2.5 sqm
Indoor total		≈ 36.3 sqm
Entry (steps, motorcycle bay, canopy)	163 × 220	3.6 sqm
Side strip along slanted fence	0–60 wide	1.4 sqm
Front yard (parking + grass)	560 / 370 / 490 / 460	21.6 sqm
Total lot		≈ 63 sqm

2nd floor — proposed		
Space	Size (cm)	Area
Room C — home office (work + gaming, bookshelves)	397 × 260	10.3 sqm
Room B — bedroom + wife's desk	380 × 220	8.4 sqm
Walk-in closet	180 × 220	4.0 sqm
Hall	500 × 120	6.0 sqm
CR / T&B (stacked over ground CR)	120 × 210	2.5 sqm
Laundry	180 × 90	1.6 sqm
Stairs	320 × 90	2.9 sqm

Key design decisions

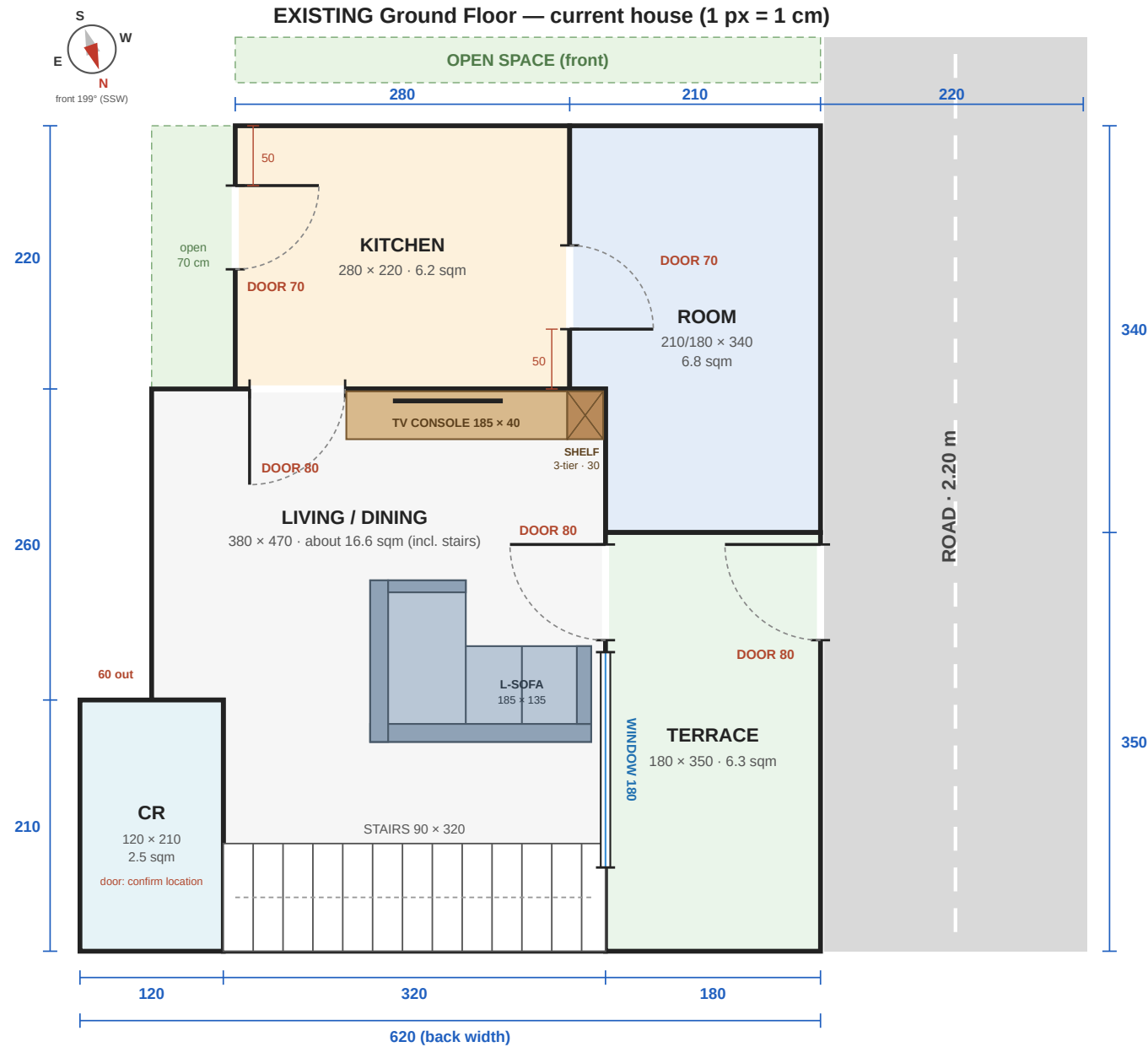
- **Living room kept as renovated** — walls, doors, TV console, shelf, sofa, stairs.
- **Kitchen at the back (north)**, dining at the front — stove smoke away from the bedroom; L-counter with exhaust hood.
- **No door to the road** — single secure entry through the gate and yard.
- **Ground-floor room kept** for parents / guests who can't use the stairs.
- **Front yard:** Jimny 5-door parking on grass pavers, grass, small tree, stepping-stone path; ADV motorcycle bay in the entry.
- **2nd floor:** CR stacked above the ground CR (shared plumbing); Room A converted to CR so the hall is bigger; walk-in closet instead of wardrobes in rooms.
- **Roof:** single slope (10°) down to the south, hidden behind a parapet for a square look; 8 × 600 W panels set back from edges; water tank under the high end above the CR.

Assumed heights (confirm with builder)

- Ground floor +0.45 m (3 steps) · floor-to-floor 3.0 m
- Roof low edge ≈ +6.45 m (front) · high edge ≈ +7.67 m (back)
- Front parapet top ≈ +7.15 m · back parapet top ≈ +8.37 m
- Windows: sill 0.90 m, head 2.10 m above floor

2 · EXISTING ground floor — current house

As built today (living room recently renovated). Use this with the proposed plans to scope demolition and new work. Stairs 320 cm; CR extends 60 cm past the living wall due to the slanted lot. CR door location to be confirmed on site.

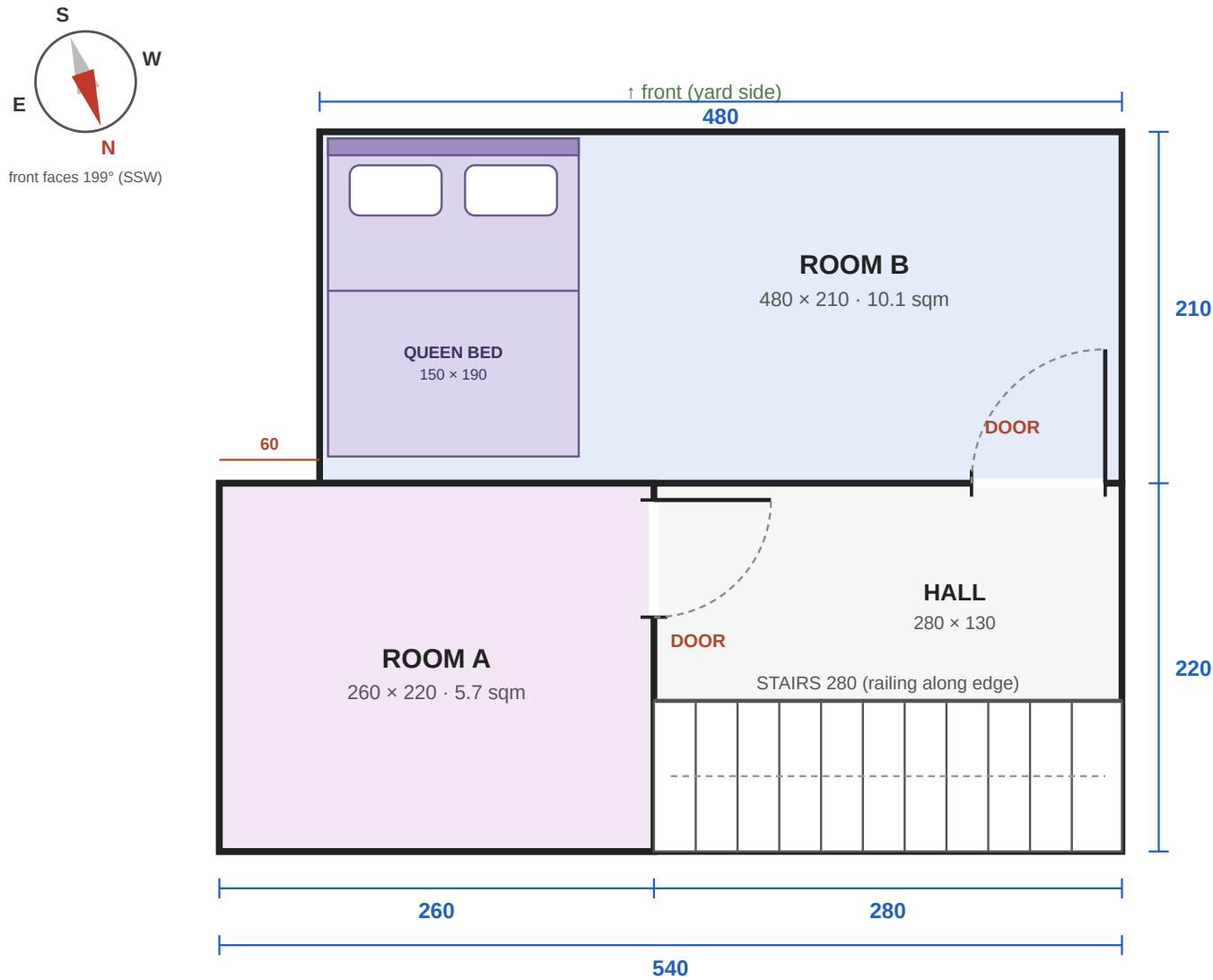


EXISTING · Indoor ≈ 32 sqm + terrace 6.3 sqm · Back 620 cm (CR 60 cm past living wall, slanted lot) · stairs 320 · furniture as renovated

3 · EXISTING 2nd floor — current house (as sketched)

From owner's sketch; exact alignment over the ground floor and existing beam to be verified on site.

2nd Floor Plan — current (1 px = 1 cm)

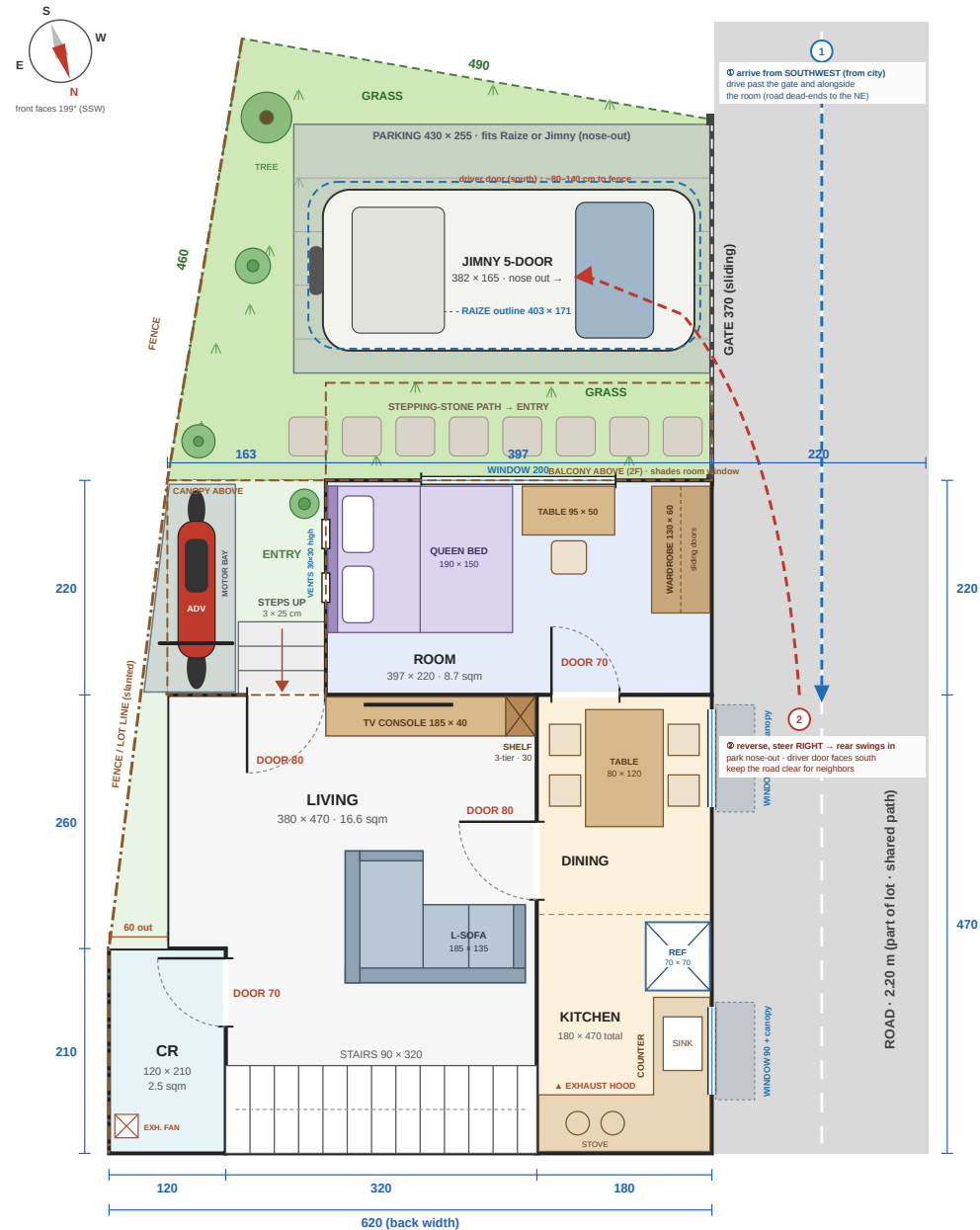


2nd floor ≈ 22 sqm (Room B 10.1 · Room A 5.7 · hall 3.6 · stairs 2.5) · No toilet or windows drawn yet

Drawn from sketch; exact alignment over the ground floor to be confirmed

Front yard, gate, parking, entry, room, living, dining/kitchen, CR · 1 px = 1 cm in the source drawing

Ground Floor Plan — Design C (1 px = 1 cm)

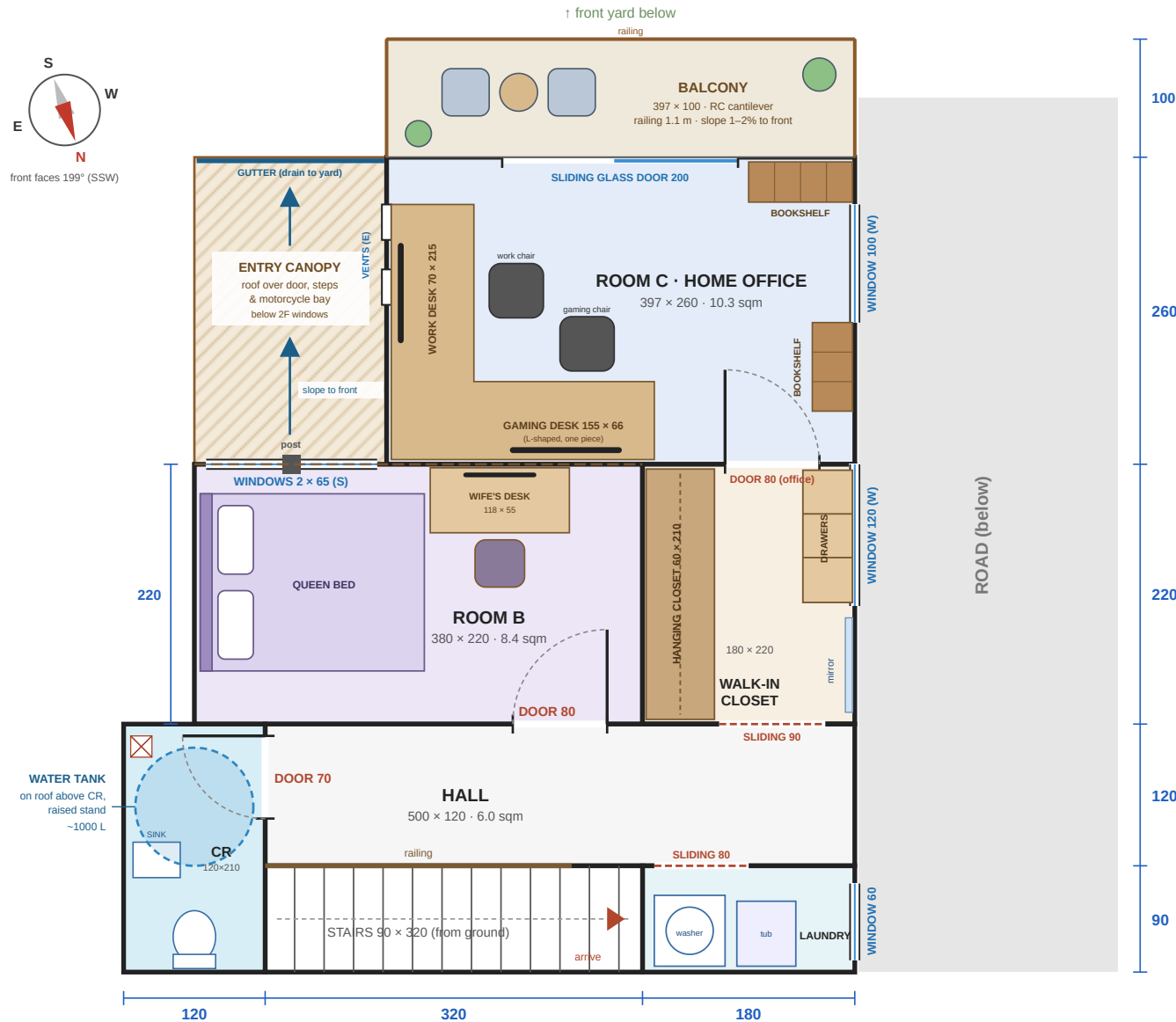


Living room unchanged · No windows on east/north (neighbors) · 2F: add window/vent at top of stairs · Front yard ≈ 21.6 sqm: parking + grass

5 • PROPOSED 2nd floor

Aligned over Design C · Room B depth fixed at 220 by existing beam · entry canopy below the 2nd-floor windows

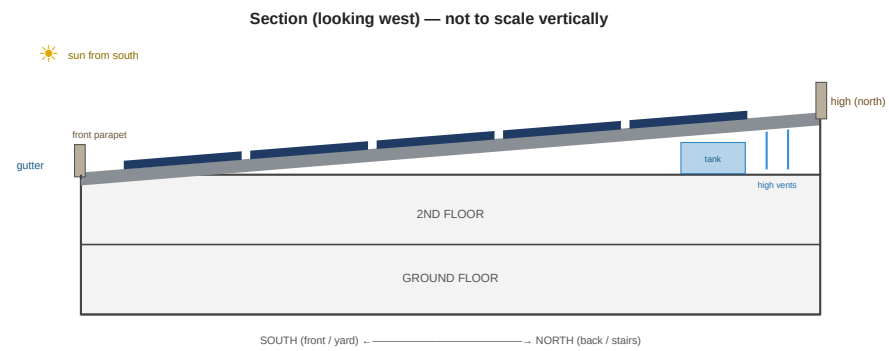
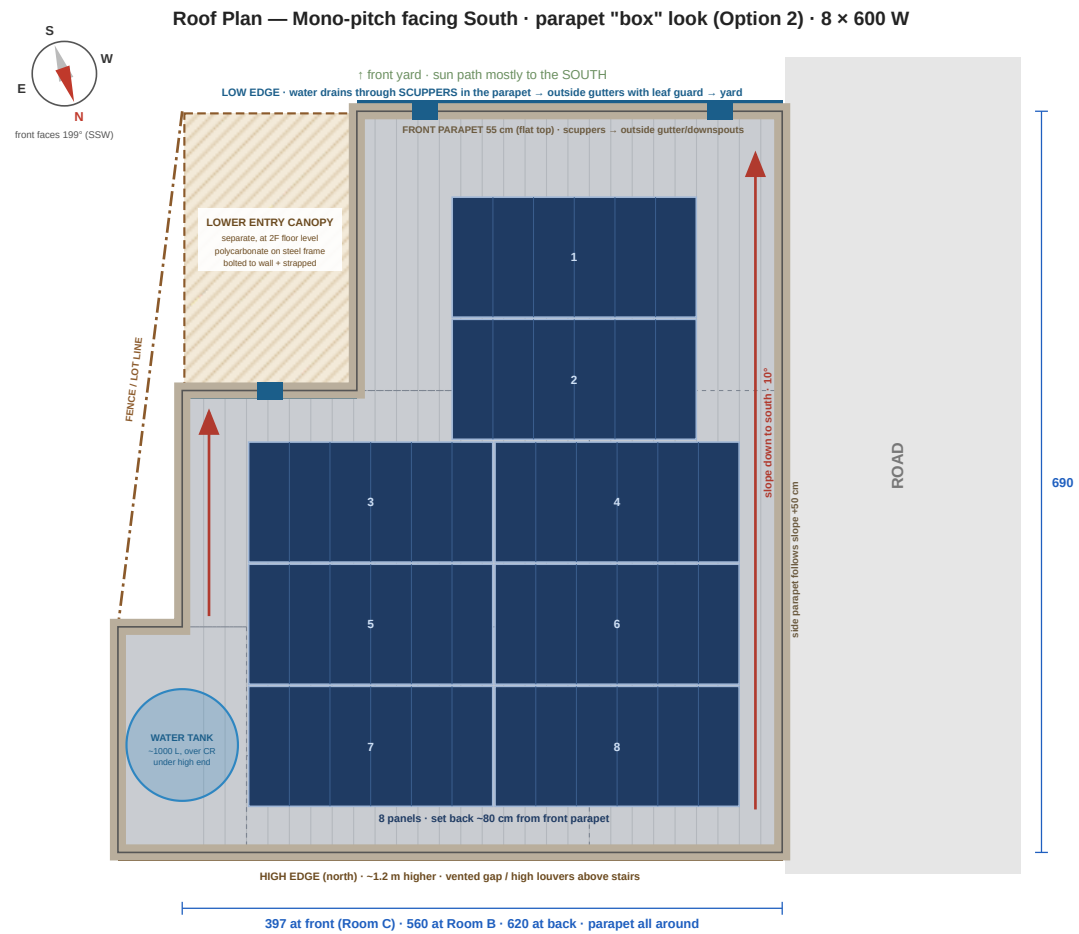
2nd Floor — Proposed (aligned over Design C, 1 px = 1 cm)



2nd floor ≈ 36 sqm (Room B 220 deep, back wall unchanged; Room C grows to 260; Room C/B wall sits on existing beam) · CR stacked over ground CR (shared plumbing)
Roof vent or turbine above stairs to pull hot air out · Structural check by engineer before extending

6 · PROPOSED roof plan & section

Mono-pitch 10° facing SSW · parapet all around (Option 2) · scuppers → outside gutters with leaf guards · 8 × 600 W panels

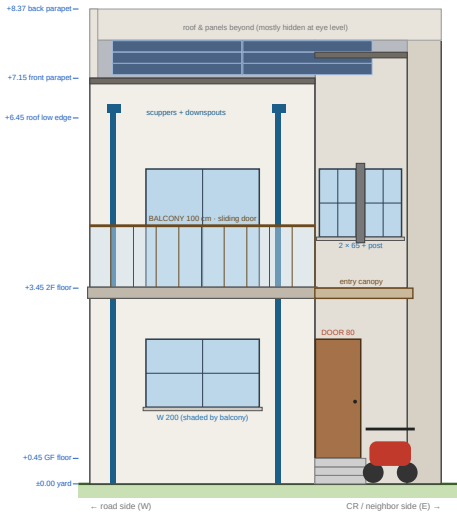


7 • PROPOSED elevations

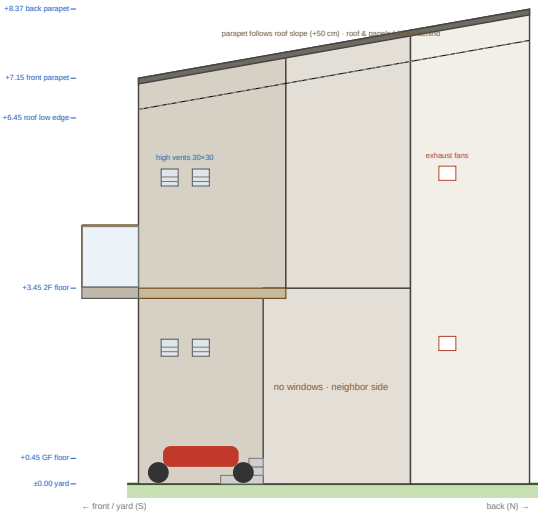
Front (SSW), right (ESE, CR side), left (WNW, road side), back (NNE)

Elevations — Design C + 2nd floor · mono-pitch roof hidden behind parapet (Option 2)

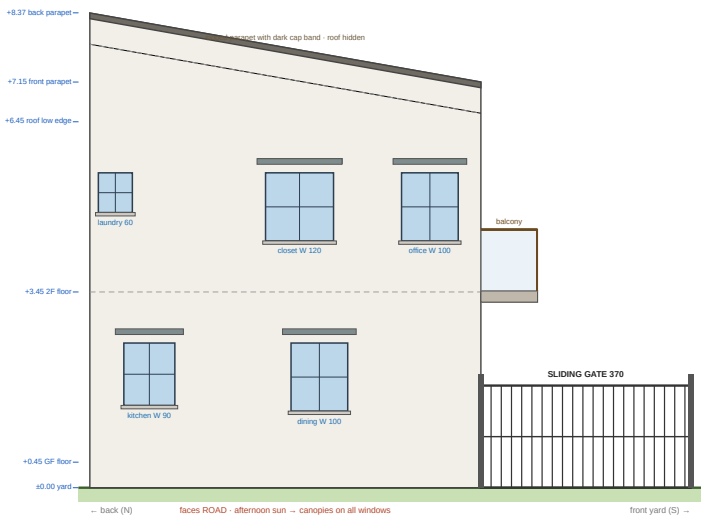
FRONT VIEW — south-southwest (facing yard, 199°)
viewer in the yard looking at the house



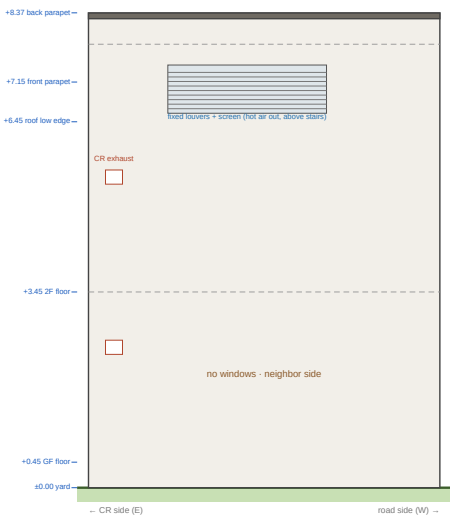
RIGHT VIEW — east-southeast (CR / neighbor side)
viewer on the right side when facing the front



LEFT VIEW — west-northwest (road side)
viewer on the road; gate and yard on the right



BACK VIEW — north-northeast
neighbor side; high end of the roof



Assumed heights: GF floor +0.45 (3 steps) · floor-to-floor 3.0 m · roof 10° rising to the back · windows sill 0.90 / head 2.10 above floor

8 • Build notes & checklist

Discuss these with your architect, structural engineer, plumber and solar installer.

Typhoon resistance (Cebu)

- Engineer to design for Cebu's wind speed per NSCP and provide wind-load calcs for roof, parapet and solar racking.
- Parapets in reinforced concrete, bars tied into beams — not plain CHB.
- Roofing ≈ 0.5 mm, tighter screw spacing at edges/corners, closer purlins at edges, straps from roof frame to ring beam.
- Solar rails bolted to purlins; panels ≥ 45 –80 cm from edges.
- High-side louvers fixed and screened — no open gaps under the roof.
- Entry canopy bolted to wall and strapped; water tank strapped to its stand.
- Typhoon-rated windows/doors on road side and front; trim trees before typhoon season.

Drainage

- Roof drains south through parapet scuppers to outside gutters and downspouts into the yard.
- Leaf guards; clean before each typhoon season. Never drain toward the east neighbor.

Solar

- 8×600 W = 4.8 kW now; roughly 16–20 kWh/day in good sun.
- 10 panels possible only with the entry roofed at main-roof level on 2 posts (drops the box look at the entry) or with larger 700 W+ panels.
- Keep the front-yard tree small; consider optimizers/microinverters if neighbors' trees shade the roof.
- Insulate under the roofing (foam / radiant barrier) — office and bedroom are directly below.

Ventilation & comfort

- Main fresh air from road side (W) and front (SSW); canopies on all road-side windows.
- Hot air exits through louvers at the high back end above the stairs.
- Exhaust fans in both CRs (no windows on east/north); exhaust hood over the stove.
- Room vents near the east property line kept small ($\approx 30 \times 30$, high) — confirm with architect (Civil Code window-distance rules).

Parents & accessibility

- Handrail on the 3 entry steps (ramp possible later).
- Grab bars and non-slip floor in ground CR; no step at CR door.
- Consider 80 cm doors for the ground room and CR.
- Night lights on the route room $\rightarrow$ dining $\rightarrow$ living $\rightarrow$ CR.

Home office

- LAN cable to the desk corner; 2–3 double outlets on a dedicated circuit.
- Split-type aircon on the west wall.

Still to confirm

- Exact lot shape (diagonals / survey vs. title).
- How the existing 2nd floor and beam align over the ground floor.
- Real floor heights and existing structure capacity for the extension.
- Canopy projection limits over the road (LGU / barangay).
- Solar panel datasheet size and inverter capacity.

9 • Owner's preliminary budget & savings plan (for discussion only)

Rough ranges from published 2026 Philippine rates plus allowances — not a contractor quote. Please price from the drawings.

Work package	Scope	Estimate (₱)
2nd floor extension	New Room C, walk-in closet, laundry (~15 sqm new)	450k – 720k
2nd floor rework	Room A → CR (plumbing, waterproofing), partitions	150k – 300k
Roof + parapet	Mono-pitch 10°, insulation, RC parapets, scuppers, gutters	300k – 550k
Ground floor rework	Terrace → kitchen/dining, kitchen+room → room, ~17 sqm	310k – 600k
Kitchen fit-out	L-counter, sink, exhaust hood	60k – 150k
Front yard	Sliding gate 370, grass pavers, grass, path, tree	100k – 250k
Entry	Canopy + handrail (or 2 posts + roof over entry for 10 panels)	25k – 90k
Windows / doors	Typhoon-rated upgrades on existing openings	50k – 150k
Water	Tank, stand, pump	25k – 60k
Solar	Remount existing 8 panels + add 2 × 600 W (inverter upgrade extra if < 6 kW)	56k – 135k
Electrical	Office circuit, LAN	20k – 60k
Professional fees	Signed/sealed plans, permits	80k – 200k
Contingency	10–15% for existing-house surprises	160k – 480k
Total		≈ 1.85M – 3.8M (mid ≈ 2.5M)

Phasing (ground floor first)

- **Phase 1** — ground floor kitchen, dining, room (semi-finished); columns/footings built for a future 2nd floor; RC slab over new areas (future 2F floor); full plans & permit once — ≈ **₱0.5–1.1M**
- **Phase 2** — replace coconut-lumber roof, 2nd-floor extension & CR, water tank, remount solar — ≈ **₱0.8–1.6M**
- **Phase 3** — finishes (tiles, paint, ceilings, cabinets), yard, gate, +2 solar panels — ≈ **₱0.6–1.2M**
- Have the roof inspected first; if termites/rot are serious, swap Phases 1 and 2 (before typhoon season).

Savings plan — ₱50,000 / month from January 2027

Funds ready for	Cumulative target (mid)	Mid	Range (low – high)
Phase 1	₱0.8M	Apr 2028 (16 mo)	Oct 2027 – Oct 2028
Phase 2	₱2.0M	Apr 2030 (40 mo)	Feb 2029 – Jun 2031
Phase 3 (complete)	₱2.9M	Oct 2031 (58 mo)	Feb 2030 – Jun 2033

Dates are when savings reach each phase's cumulative cost; construction (≈2–4 months per phase) follows. Prices usually rise yearly — keep a buffer. A Pag-IBIG home improvement loan could shorten Phase 2.

Excluded

- Furniture, aircon units, appliances, solar battery
- Structural repairs found after opening walls/roof

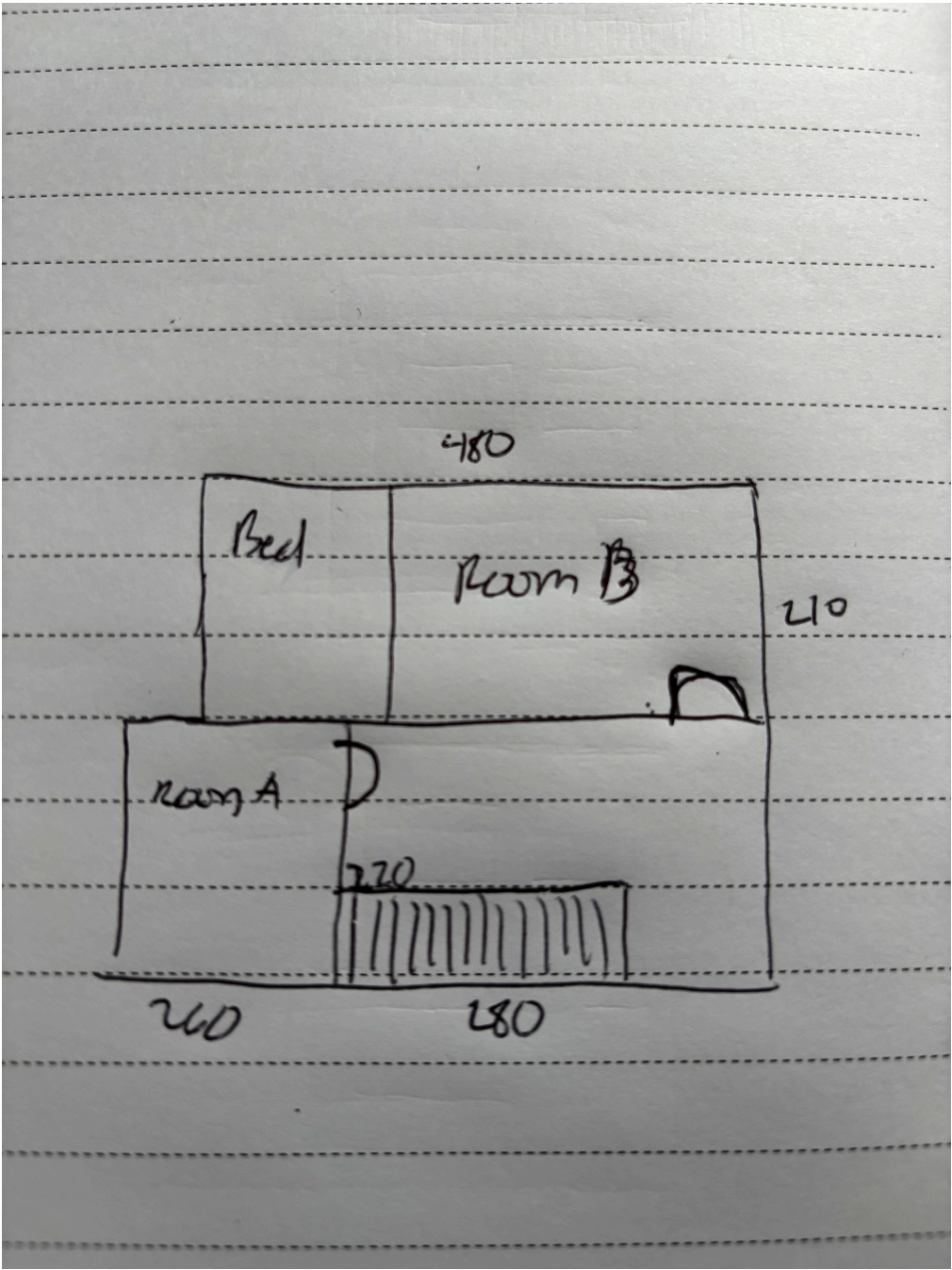
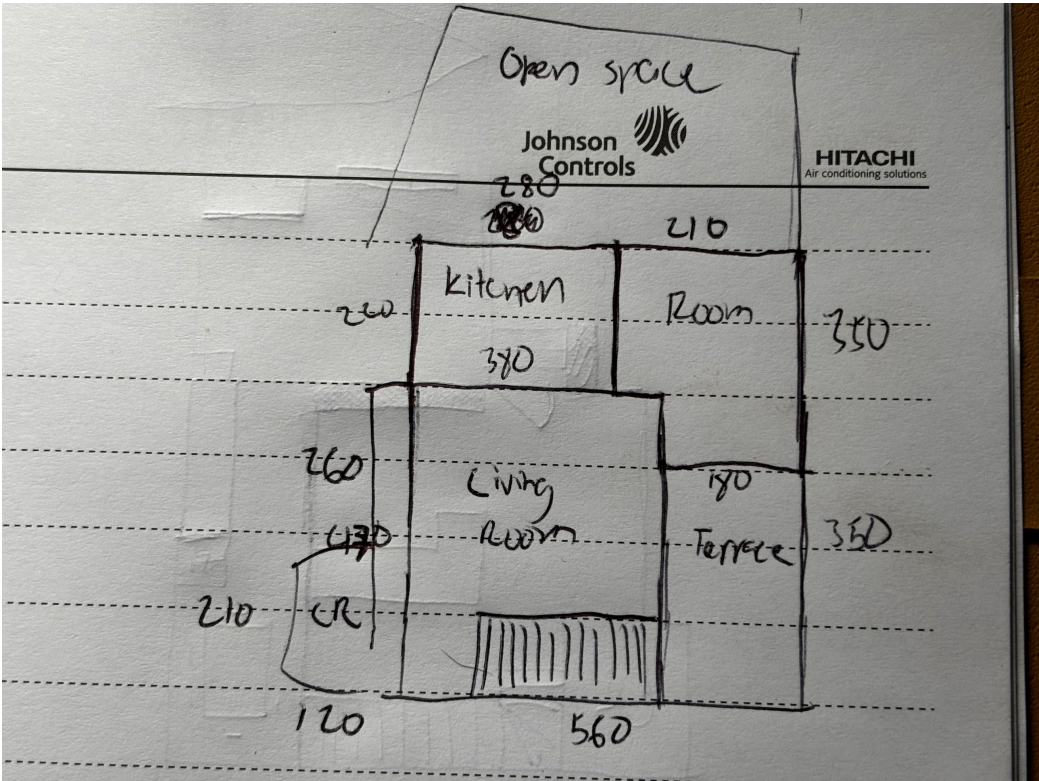
Reference rates used

- New build ₱27k–45k/sqm (standard to mid-range)
- Extension / full renovation ₱18k–35k+/sqm
- 600 W panel ₱8k–15k; installed ₱36k–56k/kW

Sources: aedoconstruction.com · jemmbuilders.com · lakasolar.ph (2026 guides)

10a · Appendix — original hand sketches

Ground floor (left) and current 2nd floor (right)



10b · Appendix — Design B (alternative explored)

Earlier option; Design C was chosen



Indoor ≈ 36 sqm · Back 620 cm (CR extends 60 cm past living wall, slanted lot) · Road-side door and other windows still to be placed